

Exercise 2: SQL Statements

Employees

id	first name	last name	department	salary	hire date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

Select Statements

- write a SQL query to retrieve all columns FROM employees table

Outcome: Employees

id	first name	last name	department	salary	hire date	city
1	John	Doe	IT	55000	2018/06/15	New York
2	Jane	Smith	HR	48000	2019/07/20	Chicago
3	Mike	Johnson	Finance	60000	2017/09/30	Los Angeles
4	Sarah	Brown	IT	53000	2021/03/25	New York
5	David	White	Marketing	52000	2016/04/10	San Francisco
6	Emily	Davis	IT	62000	2015/02/14	Chicago
7	Robert	Wilson	Finance	59000	2019/10/01	Houston
8	Jessica	Moore	HR	51000	2018/05/22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022/06/01	Chicago
10	Laura	Hall	IT	50000	2020/08/10	San Francisco

2. SELECT distinct Statement

• write a SQL query to find all unique departments in the employees table

SELECT distinct department

FROM employees;

Outcome	Department
	IT
	HR
	Finance
	Marketing

3. ORDER BY Statement

• write a SQL query to retrieve all employees first and last names ordered by salary in descending order.

SELECT first_name, last_name
FROM employees

ORDER BY salary DESC;

Outcome	First-name	Last-name
1	Emily	Davis
	Mike	Johnson
	Robert	Wilson
	John	Doe
	Sarah	Brown
	Daniel	Clark
	David	White
	Jessica	Moore
	Laura	Hall
	Jane	Smith

• Last Statement

• write a SQL query to retrieve the top 5 highest-paid employees

Select *

from employees

order by salary desc

limit 5;

outcome	First name	Last name	Salary
	Emily	Davis	62,000
	Mike	Johnson	60,000
	Robert	Williams	59,000
	John	Doe	55,000
	Sarah	Brown	53,000

5. WHERE Statement

• write SQL query to find employees who work in the IT department

SELECT *

FROM employees

WHERE department = 'IT';

outcome	First name	Last name
	John	Doe
	Sarah	Brown
	Emily	Davis
	Laura	Hall

6. AND Statement

write a SQL query to find employees who work in Finance department AND have a Salary greater than 58000

SELECT*

FROM employees

WHERE department = 'Finance'

AND Salary > 58000

Outcome	First name	Last name	Salary
	Mike	Johnson	60000
	Robert	Wilson	59000

7. OR Statement

write a SQL query to find employees who work in the HR department OR marketing department

SELECT*

FROM employees

WHERE department = 'HR'

OR department = 'Marketing';

outcome HR or marketing employees

First name	Last name
Jane	Smith
Jessica	Moore
David	White
Daniet	Clark

1. SQL Statement

write a SQL query to find employees who do not work in IT department

Select *

From employees

Where Not department = 'IT';

output employees not in IT

First name	Last name
June	Smith
Mike	Johnson
David	White
Robert	Wilson
Jessica	Moore
Daniel	Clark

2. IN Statement

* write a SQL query to find employees who are in the HR, IT or Finance department

SELECT *

FROM employees

WHERE department IN ('HR', 'IT', 'Finance');

First name	Last name
John	Doe
Jane	Smith
Mike	Johnson
Sarah	Brown
Emily	Davis
Robert	Wilson
Jessica	Moore
Laura	Hall

• Salary Condition

write SQL query to find employees who are in IT department, have a salary greater than 50000 and located in New York

Select

FROM employees

WHERE department = 'IT'

AND Salary > 50000

AND City = 'New York';

Outcome Employees with Salary > 50 000 in New York

First name	Last name	Salary	City
John	Doe	55000	New York
Sarah	Brown	53000	New York

ii. Combining WHERE, AND and ORDER BY

• Write a SQL query to find employees who work in Finance or Marketing department, earn more than 52000 and order the results by ~~Salary~~ Salary in descending order

SELECT first-name, Last-name

FROM employees

WHERE department IN ('finance', 'Marketing')

AND Salary > 52000

ORDER BY Salary DESC

Outcome	First name	Last name	Salary
	Mike	Johnson	60000
	Robert	Wilson	59000
	Daniel	Clark	53000

12. Combining SELECT distinct, WHERE and IN

* write a SQL query to find the unique cities where employees work excluding those in the IT and HR department

```
SELECT Distinct city  
FROM employees  
WHERE department NOT IN ('IT', 'HR');
```

Outcome	City
	Los Angeles
	San Francisco
	Houston
	Chicago

13. Combining WHERE, NOT, AND and ORDER BY

* write a SQL query to retrieve employees who are NOT in the finance department have a salary greater than 50000 and order the results by hire date in ascending order

```
SELECT *  
FROM employees  
WHERE department <> 'finance'  
AND salary > 50000  
ORDER BY hire-date ASC;
```

Outcome	First name	Last name	Hire date
	David	White	2016-04-10
	John	Doe	2018-06-15
	Jessica	Moore	2018-05-22
	Sarah	Brown	2021-03-25
	Daniel	Clark	2022-06-01

Combining WHERE, OR, IN and Limit

Write SQL query to find the first 3 employees who work in either Chicago or Los Angeles and belong to the IT or marketing department

SELECT

FROM employees

WHERE city IN ('Chicago', 'Los Angeles')

AND department IN ('IT', 'Marketing')

LIMIT 3;

Outcome	First name	Last name	department	City
	Emily	Davis	IT	Chicago
	Daniel	Clarke	Marketing	Chicago